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Problem Statement : 20

Project Title: Hospital Bed & ICU Allocation
Dashboard

Primary Domain: Healthcare & Telemedicine

Target Actors: Triage Nurse, Hospital
Administrator

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Lab 1 - Requirements Table

Hospital Bed & ICU Allocation Dashboard

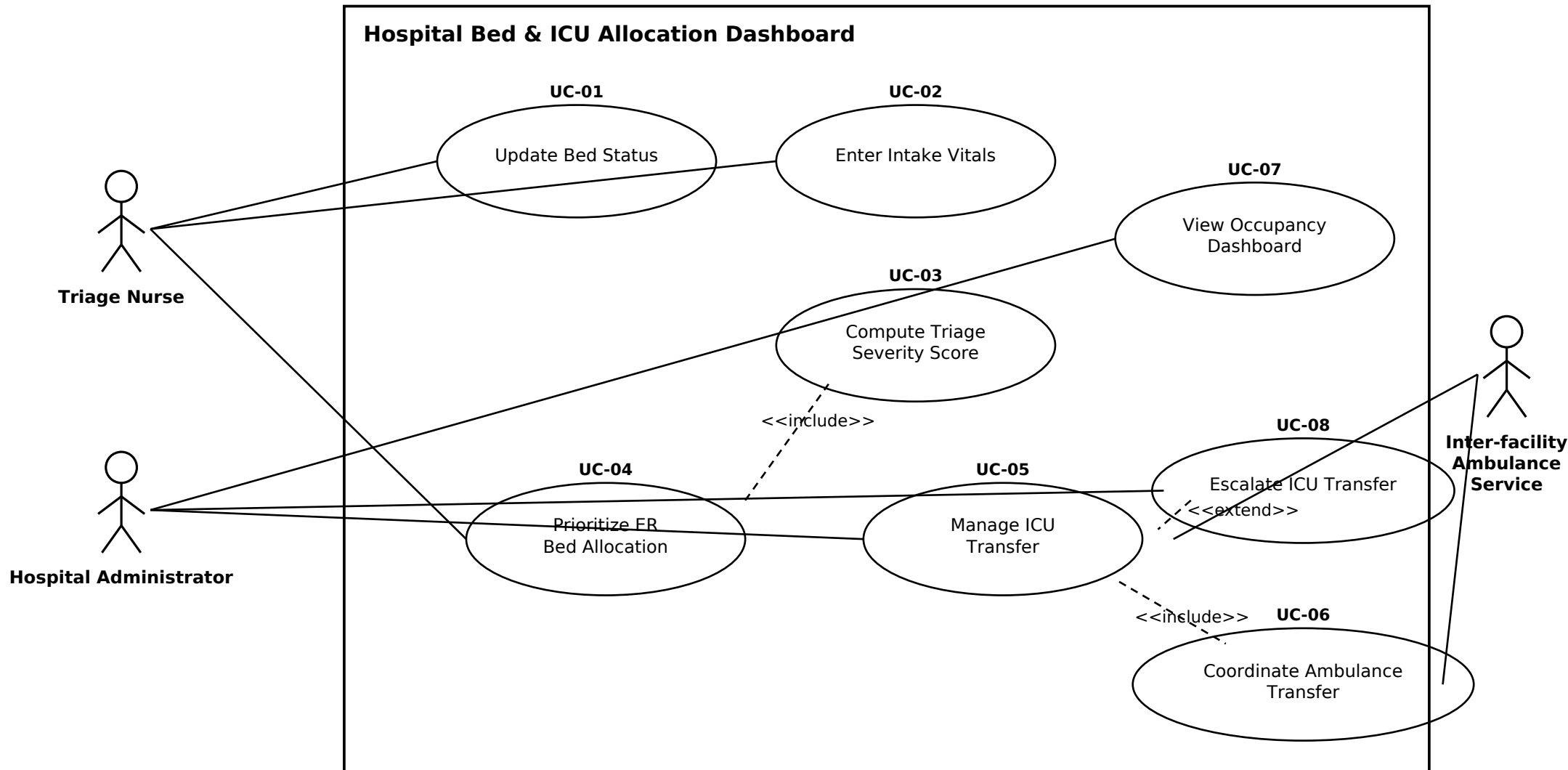
Scenario basis: Emergency room bed management dashboard with real-time ward occupancy, triage-based prioritization, and inter-facility ambulance transfer workflows.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute triage severity scores from intake vitals and dynamically order the ER bed allocation waitlist.	High	Pass: A higher-acuity patient is placed ahead of a lower-acuity patient when both are waiting. Fail: FIFO order leaves the higher-acuity patient behind.	Ensures emergency cases receive priority based on triage severity.
FR-002	Functional	The system shall display current bed status for monitored wards using Occupied, Cleaning, and Available states.	High	Pass: When a bed status changes, the new state is shown on the connected ER dashboard. Fail: The dashboard shows a previous state after the update.	Provides current occupancy information for bed allocation decisions.
FR-003	Functional	The system shall allow a Triage Nurse to enter patient intake vitals and use them to compute the patient's triage severity score.	High	Pass: After valid intake vitals are submitted, a triage score is produced and attached to the patient record. Fail: No score is produced from a completed intake.	Supplies the clinical input required for severity-based prioritization.
FR-004	Functional	The system shall prioritize patients requiring ICU transfer based on their triage severity score and identify them for transfer handling.	High	Pass: When a patient meets the configured ICU-transfer priority condition, the patient is flagged for ICU transfer handling. Fail: A qualifying patient remains unflagged.	Connects triage prioritization with ICU transfer workflow.
FR-005	Functional	The system shall create and track an inter-facility ambulance transfer workflow from transfer request through transfer confirmation.	Medium	Pass: A transfer record shows request status and is updated when transfer confirmation is received. Fail: The transfer cannot be tracked to a confirmation state.	Supports the required inter-facility ambulance coordination process.
NFR-001	Nonfunctional	The system shall propagate bed status transitions (Occupied, Cleaning, Available) to all connected ER dashboard clients in under 1 second via WebSockets.	High	Pass: Benchmarking tests show propagation below 1 second under the simulated peak-load test. Fail: Propagation reaches or exceeds 1 second.	Keeps the dashboard sufficiently real-time for bed management.
NFR-002	Nonfunctional	The system shall allow only authorized users to update bed status, triage information, or transfer workflow records and shall log rejected access attempts.	High	Pass: An unauthorized update is rejected and a corresponding access event is recorded. Fail: An unauthorized user can modify protected records or no rejection event is logged.	Protects operational and patient-related data from unauthorized changes.

Actors reflected in the UML: Triage Nurse, Hospital Administrator, and Inter-facility Ambulance Service. The first two are the stakeholders/actors named in the problem statement; the ambulance service is represented as an external actor because the scenario explicitly includes inter-facility ambulance transfer orchestration.

Lab 1 - UML Use-Case Diagram

Hospital Bed & ICU Allocation Dashboard



Relationships: `<<include>>` = mandatory reused behavior; `<<extend>>` = optional or conditional behavior.

Actors: Triage Nurse, Hospital Administrator, Inter-facility Ambulance Service.

Lab 1 - Use-Case Flow Specification

UC-04: Prioritize ER Bed Allocation

Primary Actor	Triage Nurse
Supporting Actor	Hospital Administrator
Preconditions	The Triage Nurse is authorized to use the dashboard; patient intake information and valid vitals are available; current bed-status information is accessible.
Postconditions	The patient is assigned an appropriate position in the ER bed allocation waitlist (or an available bed is allocated), the decision is recorded, and the dashboard reflects the current allocation state.

Main Success Scenario

1. Triage Nurse selects the bed allocation function for a patient awaiting ER placement.
2. The system retrieves the patient intake vitals and current bed availability information.
3. The system computes the patient's triage severity score from the intake vitals.
4. The system compares the patient's severity with the severity of patients already on the ER waitlist.
5. The system dynamically orders the waitlist so higher-acuity patients are ahead of lower-acuity patients.
6. If a suitable ER bed is available, the system associates the patient with that available bed; otherwise the patient remains in the ordered waitlist.
7. The system records the allocation/waitlist decision and the relevant triage score.
8. The updated allocation and bed status are propagated to connected ER dashboard clients.
9. The use case ends successfully.

Alternate Flow - No Suitable ER Bed Available

- 6a1.** At Step 6, no suitable ER bed is available for the patient.
- 6a2.** The system keeps the patient in the dynamically ordered ER waitlist according to triage severity.
- 6a3.** The dashboard displays the patient's waitlist status for the authorized user.
- 6a4.** If the patient requires ICU transfer handling based on the triage priority, the patient is flagged for the ICU transfer workflow; the ambulance transfer workflow can then be initiated when inter-facility transfer is required.
- 6a5.** The use case returns to the main flow at Step 7.